

Lens Patent Dataset Overview

What we loaded, how we cleaned it, and what's in it

Mantis IP Visualization

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1 What is the Lens Dataset?

Lens.org is a free, open-access patent search platform maintained by Cambia, a non-profit organization focused on democratizing access to innovation. It aggregates patent records from major offices worldwide—including the USPTO, EPO, WIPO, JPO, and CNIPA—covering over 150 million patents and patent applications. Each record includes structured bibliographic metadata, legal status, family linkages, classification codes, and, where available, full text fields such as the abstract, description, and claims. The Lens API allows bulk export of structured JSON records, making it well-suited for downstream machine learning tasks.

2 What’s in the raw dataset

The raw export contains **8,357 records** drawn from the Lens database via a targeted query on IP relevant to the Mantis visualization project. Records span 43 distinct jurisdictions and 20 languages, and cover publications from the late 19th century through 2024, though the overwhelming majority were published after 2017. The dataset is intentionally broad and largely uncleaned at this stage.

2.1 Jurisdictions and languages

The raw pull spans 43 jurisdictions and 20 languages. Non-US records make up the majority, with China, Japan, and the EPO being the largest non-US sources. The multilingual character is a direct consequence of the broad query scope and is resolved during cleaning (Section 3).

2.2 Field completeness

Text field coverage is uneven. The abstract is the most reliably present field, while description and claims are missing from over half of records—typical for non-US or older filings. The sequence listing field is not semantically relevant and was dropped during cleaning.

2.3 Publication types and legal status

The dataset is a mix of applications and granted patents, with a long tail of less common types. Most patents are either active or pending, reflecting the recency of the dataset.

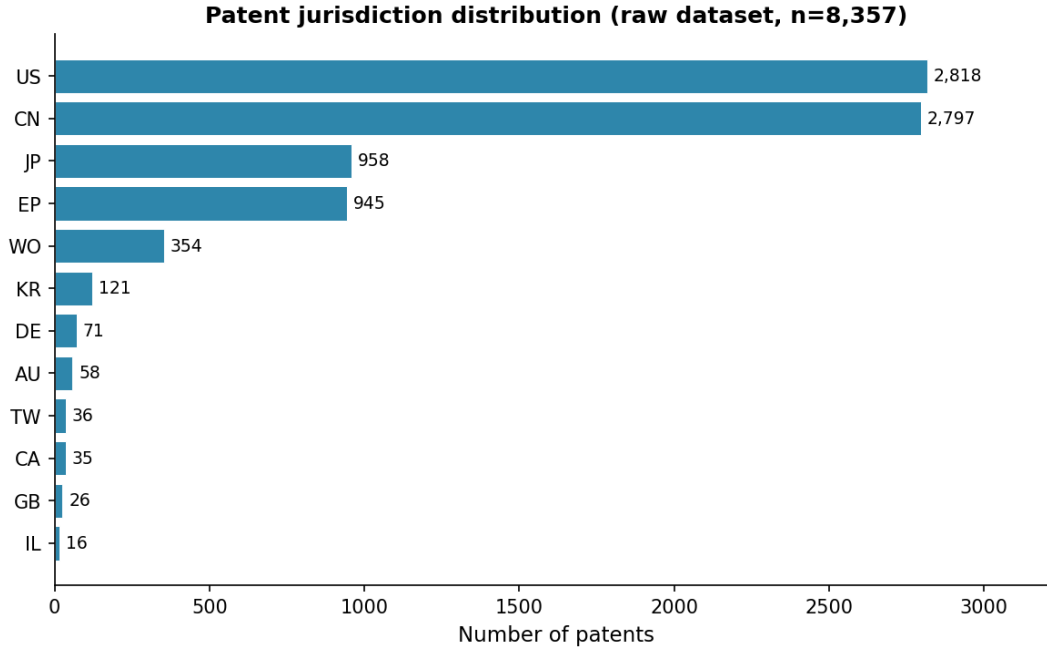


Figure 1: Jurisdiction distribution across the raw dataset (top 12 offices shown, $n = 8,357$).

2.4 Publication year

The dataset is heavily skewed toward recent filings, with the bulk of records from 2020 onward. There is a thin tail of historical records extending back to the 19th century.

2.5 Technology coverage (IPC sections)

All eight WIPO top-level sections are represented. The heaviest concentration is in Physics (G) and Electricity (H), consistent with the technology domains most relevant to Mantis.

3 How we cleaned the data

The raw export was too broad and inconsistent for direct use in Mantis. Three sequential filters were applied, followed by a structural flattening of the nested JSON schema.

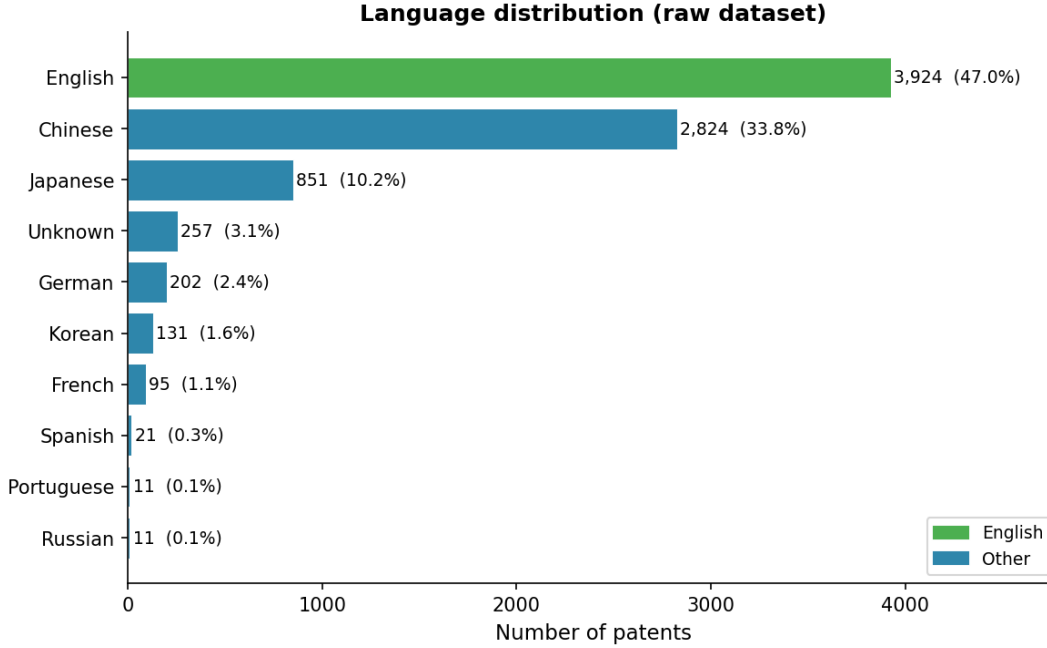


Figure 2: Language distribution across the raw dataset.

3.1 Filtering steps

1. **US jurisdiction only.** Records from non-US patent offices were removed. This reduced the dataset from 8,357 to 2,818 records and incidentally resolved the multilingual issue: after this filter, 99.9% of remaining records are in English, with three records carrying no language tag (presumed English given US jurisdiction).
2. **Abstract required.** Records with no abstract field were dropped. The abstract is the primary semantic signal used by Mantis; records without one are not useful. After the jurisdiction filter, only 39 of the 2,818 US records lacked an abstract, leaving 2,779.
3. **Deduplication.** Records were deduplicated on `lens_id`. No duplicates were found in the US-filtered subset, so the count remained at 2,779.

3.2 Schema flattening

The raw Lens records use a deeply nested JSON structure (e.g., inventor names nested inside `biblio>parties>inventors>extracted_name>value`). All records were flattened to a single-level schema with 19 columns. Fields that were either redundant, internal database identifiers, or not relevant for patent identification or semantic analysis were dropped during this step. Any column missing from a given record was filled with the string "null" to ensure a consistent schema across all 2,779 samples, as required by Mantis.

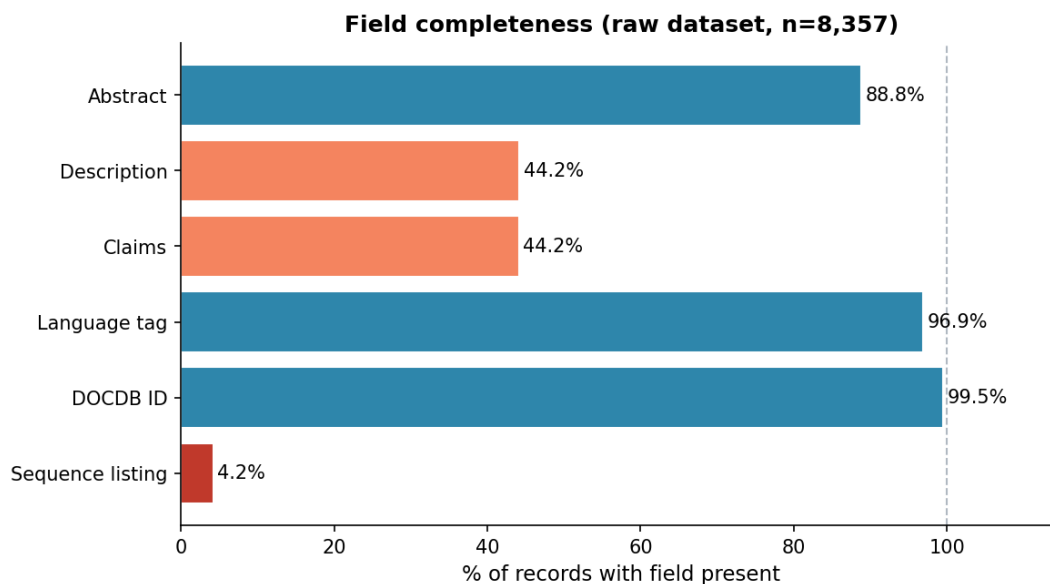


Figure 3: Percentage of raw records with each field populated.

4 Final dataset at a glance

Property	Value
Total records	2,779
Jurisdiction	US only
Language	English (99.9%), unknown (0.1%)
Fields per record	19
Records with abstract	2,779 (100%)
Records with description	1,196 (43.0%)
Abstract: mean length	105 words
Abstract: median length	107 words
Abstract: 90th pctl	149 words

Table 1: Summary statistics for the cleaned dataset.

5 Schema reference

Table 2 describes each of the 19 columns in the cleaned dataset.

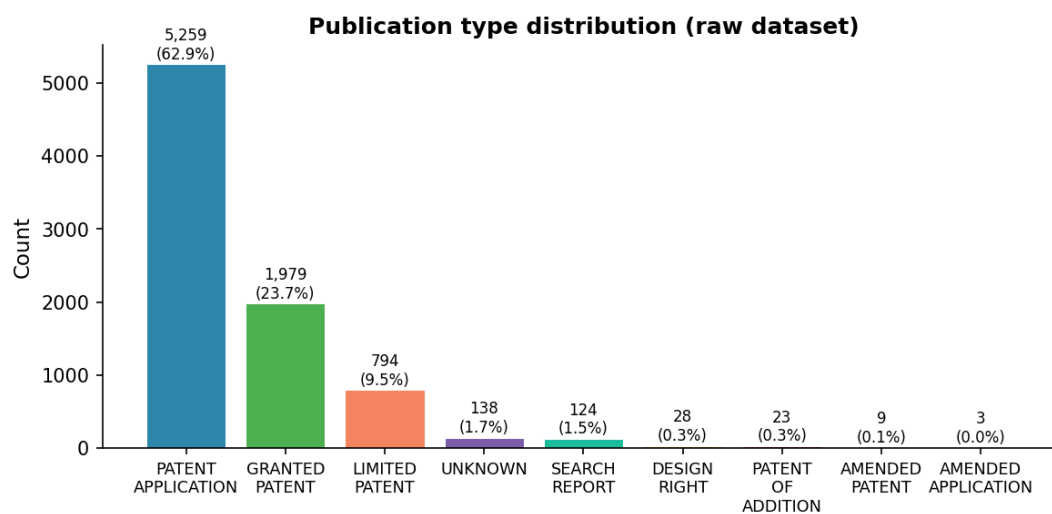


Figure 4: Publication type distribution across the raw dataset.

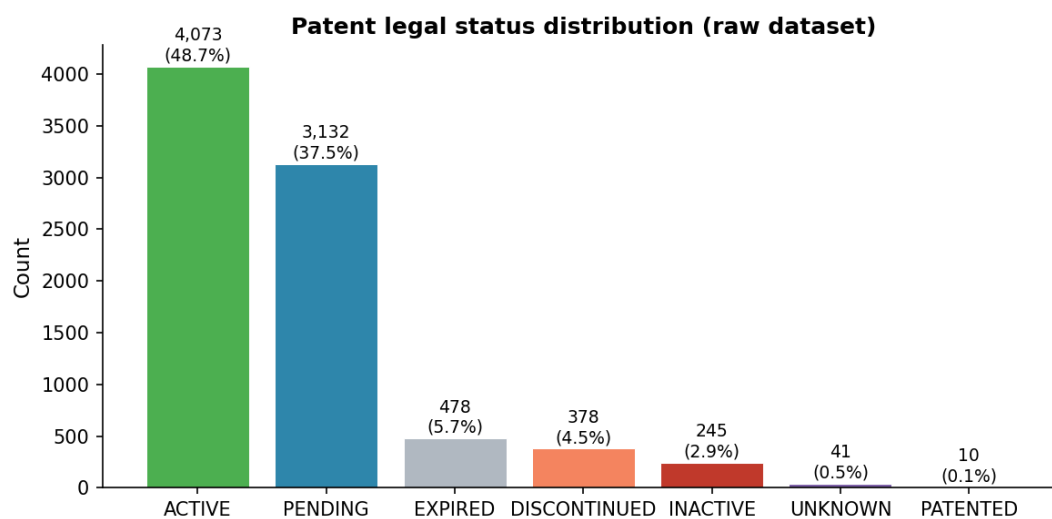


Figure 5: Legal status distribution across the raw dataset.

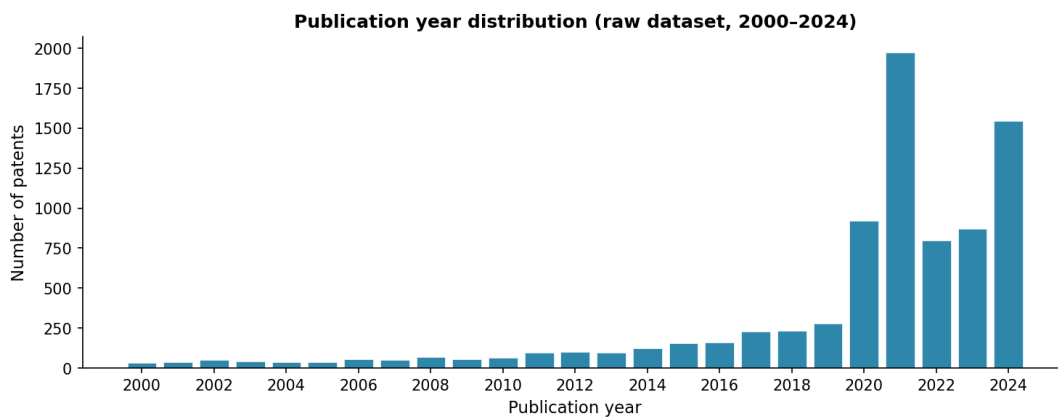


Figure 6: Publication year distribution (raw dataset, 2000–2024).

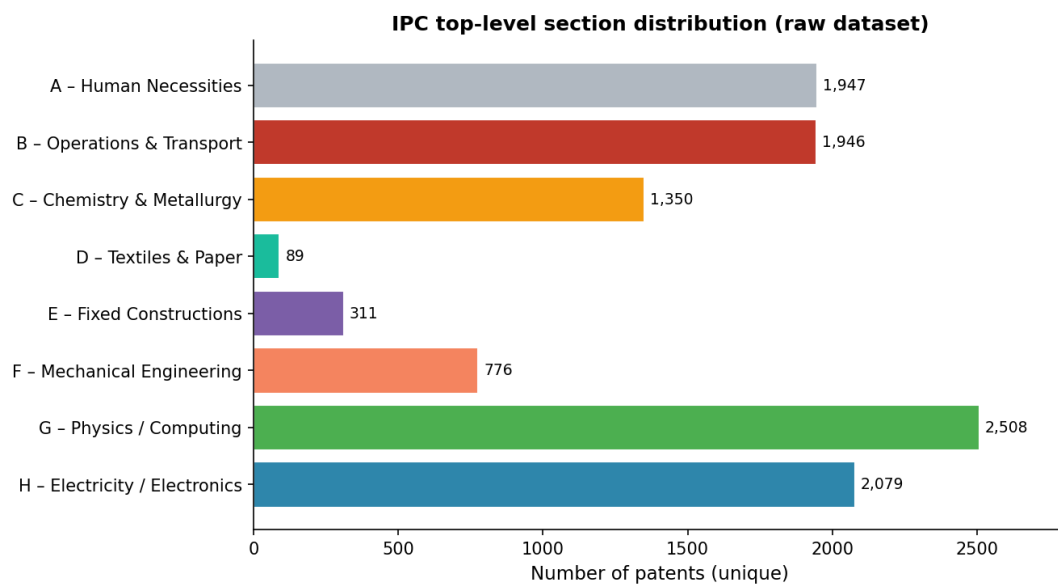


Figure 7: Number of unique patents per IPC top-level section. A patent may appear in multiple sections.

Field	Description
lens_id	Globally unique Lens.org identifier for the patent record.
jurisdiction	Patent office jurisdiction (always US in this dataset).
doc_number	USPTO document number.
kind	Document kind code indicating publication stage (e.g., A , B2).
date_published	Date the document was published (YYYY-MM-DD).
lang	Language of the document text.
publication_type	Lens-assigned publication category (e.g., PATENT_APPLICATION , GRANTED_PATENT).
patent_status	Current legal status (e.g., ACTIVE , PENDING , EXPIRED).
title	Invention title(s) as a list; may include translations.
application_date	Date the application was filed with the USPTO.
priority_date	Date of the earliest priority claim.
applicants	List of applicant (assignee) names.
inventors	List of inventor names.
ipc_classifications	List of IPC classification symbols assigned to the patent.
cited_by_lens_ids	Lens IDs of patents that cite this record (forward citations).
simple_family_lens_ids	Lens IDs of members of the simple patent family.
extended_family_lens_ids	Lens IDs of extended patent family members.
abstract	Full abstract text as filed.
description	Full description text, where available.

Table 2: Schema of the cleaned dataset (19 fields). Fields with no value for a given record are stored as the string "null".